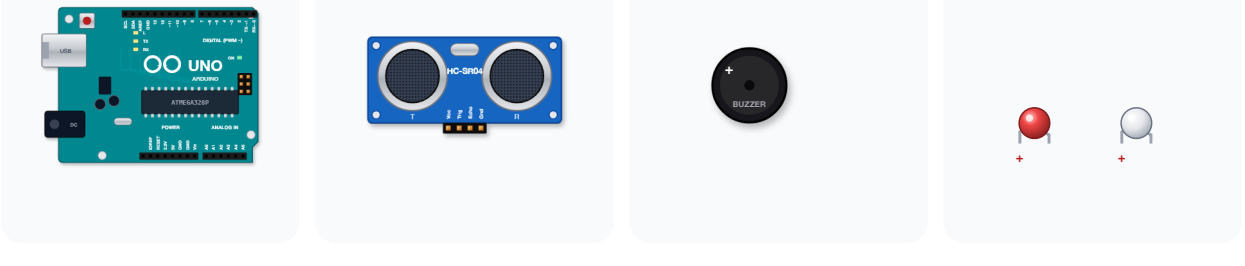
MODULE 07

07

Warm-up Lessons

Use the free Learn Arduino website to learn every idea the car's code uses, then test it on a real UNO.



Name _____ Class / Group _____ Date _____

MODULE 07

1 Warm-up with the Learn Arduino website

learn-arduino-red.vercel.app is a free website with 16 interactive 3D lessons for the Arduino UNO. Each lesson shows the board and parts in 3D with the wires drawn. The **Overview, Pinout, Example and Specifications** tabs explain it, and you can **run the sketch in the browser** and see the result. Do the lessons marked **Must do** before you build the car: they teach every idea the car's UNO code uses.

LEARNING OBJECTIVES · ABOUT 90 MIN

By the end of this module you can:

- Explain digitalWrite, analogWrite (PWM), pulseIn and map() in your own words
- Read an HC-SR04 distance and turn it into zones with comparisons
- Move a sketch from the website's pins to the robot car's pins and test it on a real UNO

How to use the website

- Open it on a computer (Chrome or Edge) for the best 3D view. Drag to move, **Q / E** to rotate, scroll to zoom.
- Click a part in the 3D scene or a lesson's controls to change its values, then press ▶ **Run sketch** and watch the serial monitor.
- The code box is **read-only**. To change it, press **Copy code**, paste it into the Arduino IDE and upload it to a real UNO.
- The website only knows the **Arduino UNO**: there is no ESP32-CAM, L298N or WiFi. Those are taught in Modules 02 and 04 of this handbook.

Lesson map: website → robot car

#	Website lesson	You learn	Where the car uses it	
1	Meet the board	The UNO's pins: digital, PWM (~), analog, power	Module 02: every wire goes to one of these pins	**Must do**
2	Blink an LED	`pinMode`, `digitalWrite`, `delay`	Red rear LED `A0`, white reverse LED `A2`, headlights `A4`	**Must do**
3	Push a button	Digital input, pull-down resistor	Not on the car (idea: a start button)	Extra
4	Turn a knob	`analogRead`, `map()`	`map()` turns distance into beep speed	**Optional**
5	Mix colours	PWM with `analogWrite` 0–255	Motor speed on ENA `D11` / ENB `D6`	**Must do**
6	Sense light	Analog sensor (LDR)	Idea: LDR on `A5` switches the headlights on when it gets dark	Extra
7	Read a sensor	HC-SR04, `pulseIn()`, ÷58 = cm	Rear sensor: TRIG `D13`, ECHO `D12`	**Must do**
8	Make a sound	Buzzer, `tone()`	Alarm buzzer on `A3`, active type, no `tone()`	**Must do**
9	Mini piano	Arrays and loops	Idea: a melody for the horn	Extra

#	Website lesson	You learn	Where the car uses it	
10	Control a motor	PWM speed + transistor driver	The L298N does the same job and also reverses the motor	**Must do**
11	Sense heat	TMP36 temperature sensor	Not on the car	Extra
12	Sweep a servo	SG90 servo, `Servo.h`	Scanning radar option `USE_SERVO`, servo on `A1`	**Optional**
13	Spin the wheel	Stepper + ULN2003	Not on the car	Extra
14	Smart plant	Sensor + decision	Same idea as auto-brake: read → decide → act	Extra
15	Feel the force	Force sensor	Not on the car	Extra
16	Build a project	**Parking sensor**: zones with comparisons	The core of our rear alarm, see Test 06	**Must do**

Same code, different pins

The website uses its own pins. The car uses other pins because the L298N, the ESP32-CAM link and the sensor all need space. **The code is the same, only the numbers at the top change.**

Part	Website pin	Robot car pin	Why the car is different
HC-SR04 TRIG / ECHO	`D9` / `D10`	`D13` / `D12`	D9 and D10 drive the L298N (IN2, IN1)
Buzzer	`D8`, passive, `tone()`	`A3`, active, `digitalWrite`	`tone()` uses Timer2 and would break the PWM on D11 (ENA)
LED	`D13` (built-in)	`A0` red, `A2` white, `A4` headlights	D13 is the sensor's TRIG on the car
Motor PWM	`~D6` → TIP120	`~D11` ENA, `~D6` ENB → L298N	Two sides of the car, and the L298N can reverse
Servo	`D9`	`A1`	Optional scanning radar
Serial Monitor	9600 baud	115200 baud	Faster, so printing does not slow the car down

Active or passive buzzer?
An **active** buzzer beeps by itself when its pin is HIGH (this is the one in our parts list). A **passive** buzzer, like the website's, needs `tone()`. On the car `tone()` would stop the left motors working properly, so keep the active buzzer.

Bridge exercise: Test 06, parking alarm on the bench

Lesson 16 lights a traffic light. Its specifications even say *"Next step: add a buzzer from lesson 8"*. `tests/06_parking_alarm_bench` does exactly that, on the robot car's pins, so the wires stay the same when you move them to the car.

From	To
HC-SR04 VCC / GND	`5V` / `GND`
HC-SR04 TRIG	`D13`
HC-SR04 ECHO	`D12`
Red LED (+) via 220 Ω	`A0`
White LED (+) via 220 Ω	`A2`
Buzzer (+)	`A3`
LED (−), buzzer (−)	`GND`

1. Wire the parts on a breadboard with the UNO only (no battery, no motors).
2. Open `tests/06_parking_alarm_bench`, select **Arduino Uno**, upload.
3. Open the Serial Monitor at **115200**.
4. Move your hand slowly towards the sensor and watch the bar grow.

Distance	Zone	Buzzer + red LED	Serial Monitor
more than 100 cm	SAFE	Off	`150 cm SAFE ##`
100 → 20 cm	WARN	Beeps, faster as you get closer (900 → 120 ms)	`55 cm WARN ######`
20 cm or less	STOP!	Continuous	`12 cm STOP! ##########`

Worksheet

#	Question	Your answer
1	Lesson 2: what does `digitalWrite(pin, HIGH)` do to the pin's voltage?	
2	Lesson 5: `analogWrite(pin, 128)`. About what percentage of full power is that?	
3	Lesson 7: the echo lasted 1160 µs. How far is the obstacle? Show your calculation.	
4	Lesson 7: why divide by 58 and not by 29? (Hint: the sound travels there *and back*.)	
5	Lesson 8: what is the difference between an active and a passive buzzer?	
6	Lesson 10: why can't an Arduino pin power a motor directly? What does the L298N add?	
7	Lesson 16: write the comparison that is true only when the obstacle is between 20 and 100 cm.	
8	Test 06: with `map(cm, 20, 100, 120, 900)`, what is the beep period at 60 cm?	

Challenges (extra)

- Add the green / yellow / red traffic light from lesson 16 on free pins and keep the buzzer working.
- Change `WARN_CM` and `STOP_CM`. What happens if STOP is bigger than WARN? Fix it in code.
- Lesson 12: mount the sensor on a servo and print the distance for 5 angles: a text radar.

Checklist

Done	Item
☐	Lessons 1, 2, 5, 7, 8, 10 and 16 run on the website
☐	Worksheet answered
☐	Test 06 works on a real UNO: SAFE / WARN / STOP! zones correct
☐	I can explain why the car's pins differ from the website's

Student _____ Teacher _____ Date _____

Learn Arduino website designed by x.com/DilumSanjaya. This module only links to it; the lessons belong to their author.